

Worst Exam Ever! (bonus)

After long weeks of waiting, Edoardo has finally got back the results of the last exam he took. The exam was made of N exercises, the i -th exercise had a maximum score of P_i points and he scored S_i points. Edoardo is not happy of the grade he received, and is considering to take it again. However, the professor doesn't want yet another exam to grade... and thus he decided to give him a generous *bonus*!

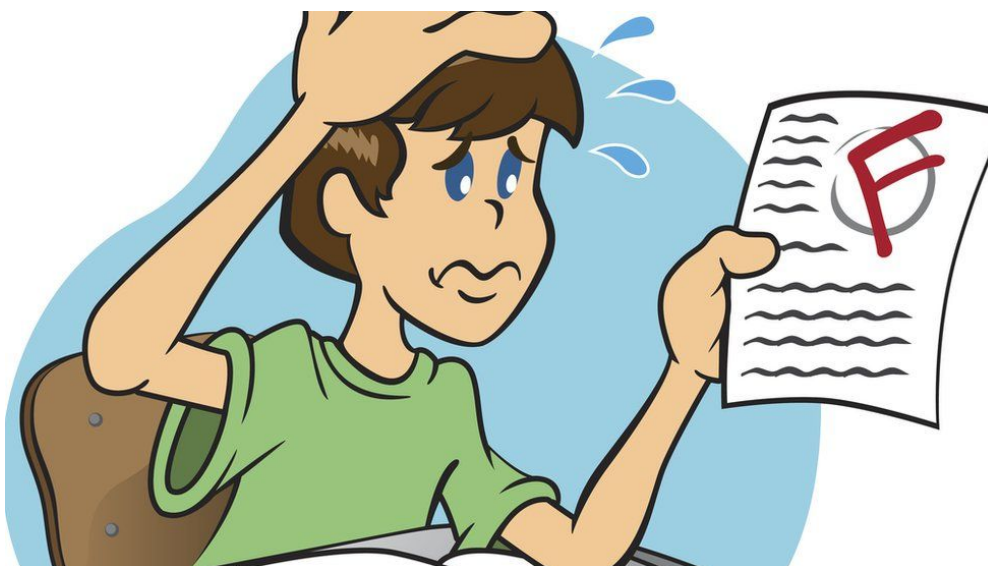


Figure 1: Edoardo is desperate!

More precisely, Edoardo can choose exactly one of the N exercises and that exercise will be discarded and will not take part in the grading of the exam. Note that the grade of the exam is defined as the ratio between the sum of points scored and the sum of maximum scores (excluding in both sums the exercise discarded, thanks to the generous bonus). Help Edoardo find which exercise to discard in order to get the best grade possible!

📎 Among the attachments of this task you may find a template file `bonus.*` with a sample incomplete implementation.

📖 Notice that you may find the wrong solution if you use floating-point variables to compare fractional values, due to precision errors. To find out if $\frac{A}{B} \geq \frac{C}{D}$ you can check instead if $A \cdot D \geq C \cdot B$, while being careful that even if the input values fit inside a 32-bit integer, the intermediate multiplications may not. Then, use 64-bit integers such as `long long` for C and C++ and `Int64` for Pascal (the templates attached already do it).

Input

The first line contains the only integer N . The following N lines contain two integers S_i, P_i each.

Output

You need to write a single line with an integer: the index (starting from zero) of the exercise to be discarded. In case more than one solution exist, print one of them.

Constraints

- $2 \leq N \leq 100\,000$.
- $1 \leq P_i \leq 10\,000$ for each $i = 0 \dots N - 1$.
- $0 \leq S_i \leq P_i$ for each $i = 0 \dots N - 1$.

Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** (0 points) Examples.

- **Subtask 2** (15 points) $N \leq 10, P_i \leq 10$ for every $0 \leq i \leq N - 1$.

- **Subtask 3** (15 points) $N \leq 1000, P_i \leq 10$ for every $0 \leq i \leq N - 1$.

- **Subtask 4** (30 points) $N \leq 5000$.

- **Subtask 5** (10 points) $P_i = P_j$ for every $0 \leq i, j \leq N - 1$

- **Subtask 6** (10 points) $S_i = S_j$ for every $0 \leq i, j \leq N - 1$

- **Subtask 7** (20 points) No additional limitations.


Examples

input	output
3 1 1 2 2 1 3	2
3 1 1 1 2 1 2	1

Explanation

In the **first sample case**, the best choice is to discard the last exercise, obtaining a grade of $\frac{1+2}{1+2} = \frac{3}{3} = 1$.
In the **second sample case**, Edoardo can choose to discard one of the last two exercises, obtaining a grade of $\frac{1+1}{1+2} = \frac{2}{3}$. This is the best grade that Edoardo can obtain.